Sugam Dahal

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Summary

Computer Engineering graduate and machine learning engineer specialising in deep learning, computer vision and large language models. Industry experience deploying a production OCR microservice built on a vision-language model with vLLM, FastAPI and Docker, and evaluating post-training quantization for efficient LLM inference. Currently first-authoring a semantic segmentation study covering 202 km^2 of the Kathmandu Valley, spanning dataset construction, architecture design and evaluation of diffusion-generated training data. Comfortable across the full applied ML workflow: data pipelines, model training and fine-tuning, experiment design, statistical evaluation and deployment.

Education

Institute of Engineering (IOE), Thapathali Campus — Kathmandu, Nepal
Bachelor of Engineering in Computer Engineering | *78.56%* — *Apr 2022 – Aug 2026*

Global School of Science — Kathmandu, Nepal
Higher Secondary Education (Science) | *GPA 3.59 / 4.00* — *Jun 2019 – Dec 2021*

Technical Skills

Programming Languages: Python, C++, JavaScript, SQL, Bash, LaTeX

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, Neural Networks, CNN, RNN, Transformers, Semantic Segmentation, Diffusion Models, Transfer Learning, Fine-Tuning, LoRA / PEFT, Model Quantization (GPTQ, AWQ), Mixed-Precision Training

LLMs & NLP: Large Language Models, GPT-2 Fine-Tuning, BPE Tokenization, Multi-Head Attention, Function Calling, Structured Outputs, Agentic Workflows, Prompt Engineering, LLM Evaluation (BLEU, Perplexity)

Computer Vision: Image Segmentation, Vision-Language Models, OCR, SegFormer, U-Net, Stable Diffusion, ControlNet, Albumentations, CVAT

MLOps & Deployment: FastAPI, Flask, vLLM, Docker, REST APIs, Git, GitHub, Linux, PostgreSQL, Weights & Biases, pytest, Pydantic, CUDA, Google Colab, Hugging Face Hub

Data Science & Statistics: NumPy, Pandas, Matplotlib, Seaborn, Data Preprocessing, Feature Engineering, PCA, Cross-Validation, Bootstrap Confidence Intervals, Experiment Design, Statistical Analysis, Data Visualization

Professional Experience

Wise Yak Solutions Pvt. Ltd. — Kathmandu, Nepal
AI Research Intern — *Dec 2025 – Mar 2026*

- Engineered a production-ready OCR pipeline built on a 1B-parameter vision-language model, extracting structured text, dense tables and mathematical notation from complex document layouts.
- Architected a dual-engine inference backend using standard PyTorch for local development and vLLM with PagedAttention for high-throughput production serving, packaged behind a FastAPI service with Docker deployment.
- Evaluated post-training quantization methods (GPTQ, AWQ) for LLM inference, comparing memory footprint and output quality to inform deployment decisions on constrained hardware.
- Investigated custom neural network architectures and algorithm optimization within the AI Research Team, supporting applied computer vision research and deployment.

Research Experience

Independent Researcher | **Land-Use Segmentation and Generative Augmentation** — Kathmandu, Nepal
Dept. of Electronics and Computer Engineering, IOE Thapathali Campus — *May 2026 – Present*

- Built and released a six-class land-use segmentation benchmark covering 202.4 km^2 of the Kathmandu Valley across 408 hand-annotated fine-resolution scenes, published on Zenodo with fold definitions, metric records and trained weights (DOI: 10.5281/zenodo.22349064).

- Designed and validated a full evaluation protocol across five architectures (U-Net, UNetFormer, SegFormer-B0, SegNeXt-T, proposed) and two data partitioning regimes, quantifying that split protocol alone shifts reported mIoU by 0.051–0.095, an effect larger than any architectural difference.
- Developed a thin-structure-aware segmentation model adding a residual detail path, boundary-reliability reweighting, a clDice topology loss and a presence-balanced sampler to a SegFormer-B0 backbone, improving on the backbone in every fold-run for 1.9% additional parameters.
- Implemented a two-stage diffusion generator (mask DDPM plus LoRA-adapted Stable Diffusion with a class-channel ControlNet) and a compute-matched downstream evaluation with bootstrap confidence intervals; first-author manuscript in preparation, preprint forthcoming.

Projects

GeoNUSAF: GeoSpatial Neural Urban Synthesis and Analysis Framework
PyTorch, SegFormer, Stable Diffusion, ControlNet, FastAPI, Weights & Biases | *Team of 4*
- Fine-tuned a SegFormer-B0 backbone for 7-class urban land-use segmentation on 512x512 satellite imagery, using class-weighted cross-entropy for imbalance, albumentations-based augmentation and 3-fold cross-validation, reaching 0.50–0.60 validation mIoU with experiment tracking in Weights & Biases.
- Built a generative urban-planning pipeline feeding segmentation masks and templated prompts into Stable Diffusion with Canny-edge ControlNet conditioning, decoupling structural guidance via a reference image, and served the system through a FastAPI backend with automated cost and time estimation.

High-Performance OCR Microservice
Python, LightOnOCR-2-1B (VLM), vLLM, FastAPI, Docker, Hugging Face Transformers
- Built an end-to-end OCR microservice around a 1B-parameter vision-language model, replacing brittle multi-stage OCR pipelines with direct image-to-text inference over PDFs, scans and images.
- Implemented configurable dual-backend inference (Hugging Face for low-VRAM development, vLLM for production throughput), aspect-preserving 200 DPI PDF preprocessing, health and diagnostic endpoints, and containerised deployment via Docker Compose.

StockSage: Autonomous Multi-Agent Financial Research Platform
LangGraph, Hybrid RAG, FastAPI, Streamlit, Docker, ChromaDB
- Built an autonomous LangGraph multi-agent system (Supervisor, Data, News RAG, Risk, Analyst, Critic) with parallel execution branches ($\sim$40% faster ingestion), SQLite state checkpointing, Human-in-the-Loop review gates, and Reflexion self-correction loops.
- Designed a multi-collection Hybrid RAG engine over SEC EDGAR 10-Q/10-K filings and live news using ChromaDB, BM25, Reciprocal Rank Fusion, and FlashRank cross-encoder reranking, achieving 100% precision on a 15-question golden evaluation benchmark.
- Shipped an async FastAPI + WebSocket backend and Streamlit/Plotly dashboard computing Wilder's RSI, 95% Historical VaR, Max Drawdown, and portfolio Pearson correlations, backed by Docker Compose and 114+ unit tests in GitHub Actions CI.

Building Large Language Models from Scratch
PyTorch, Transformers, Multi-Head Attention, BPE Tokenization, tiktoken
- Implemented a complete GPT-style language model from first principles across 11 progressive modules: BPE tokenization, sliding-window data loaders, token and positional embeddings, multi-head self-attention, causal masking, layer normalisation and cross-entropy training with perplexity tracking.
- Engineered a weight-mapping pipeline converting pretrained OpenAI GPT-2 (124M) TensorFlow checkpoints into the custom PyTorch architecture for verified text generation and inference.

Interactive Chatbot for Mental Wellbeing
Python, PyTorch, GPT-2 Fine-Tuning, Flask, PostgreSQL, React | *Team of 4*
- Fine-tuned GPT-2 on 6 domain-specific Hugging Face mental health datasets over 10 epochs (batch size 4, learning rate 3e-5) to generate contextually aware supportive responses, evaluating generation quality with BLEU score.
- Built the Flask REST API and a four-table PostgreSQL schema handling authentication, interaction logging, ratings and feedback capture, integrated with a React frontend for real-time conversation.

Network Intrusion Detection System (NSL-KDD)
Python, Scikit-learn, XGBoost, Pandas, PCA
- Built an automated ML pipeline for network intrusion detection featuring programmatic Kaggle dataset extraction, categorical encoding, robust feature scaling and optional PCA dimensionality reduction.
- Benchmarked 7 classifiers (XGBoost, Random Forest, Linear SVM, KNN, Naive Bayes, Decision Tree, Logistic Regression) on accuracy, precision and recall with automated confusion-matrix and feature-importance visualisations.

Certifications

The Complete Full-Stack Web Development Bootcamp – Udemy, 2024

Hardware Fellowship Program – Locus, 2023

Research Interests

Deep learning with a focus on computer vision, large language models and NLP, and generative models, with growing interests in reinforcement learning, mechanistic interpretability, and representation learning. My goal is to understand not just how these systems perform, but why they work—whether leading to theoretical insights or deployable intelligent systems built with empirical rigor.